

Mikael Gaete López

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PROFILE

Full-stack developer and Computer Engineering student (Universidad de Chile), with experience spanning backend development and databases to network security and cloud infrastructure. I have led teams, migrated mission-critical production systems with zero data loss, and I currently research the application of prompt engineering techniques to improve SQL query translation systems.

WORK EXPERIENCE

Thesis Researcher

Mar 2026 — Dec 2026

Department of Computer Science, Universidad de Chile

Santiago, Chile

- Integrated a mathematical abstraction framework into an SQL query translation system (built on static analysis tools and artificial intelligence) using Prompt Engineering techniques to improve the translation process between database engines.
- Increased the semantic equivalence of the translations produced by the SQL query translation system across database engines through advanced prompt design via In-Context Learning.
- Incorporated agentic coding tools into the workflow, minimizing time spent on repetitive tasks and allowing focus on the most relevant areas of the work.

Tech Lead

Mar 2026 — Jul 2026

Department of Computer Science, Universidad de Chile

Santiago, Chile

- Designed and developed a web platform that automated the assignment of medical staff and consultation rooms for a public primary care facility, replacing a fully manual process and reducing assignment time by up to 75%.
- Led a team of 5 developers under an agile methodology (using Jira), with structured code reviews, biweekly meetings, and a GitHub branching strategy that minimized integration conflicts.
- Incorporated agentic coding tools into the team's workflow under a spec-driven approach: defining requirements and interface contracts before any generation, using artificial intelligence to accelerate implementation without delegating architecture decisions or code review.
- Tech stack: React, TypeScript, and TailwindCSS on the front end; Node.js, Express, and TypeScript on the backend; and PostgreSQL together with Prisma ORM for the database.

Full-Stack Developer

Feb 2026 — Apr 2026

Freelance

Santiago, Chile

- Developed a content management module that enables creating and publishing dynamic pages entirely from the admin panel, eliminating the need for technical intervention for new content.
- Implemented file upload, storage, and distribution (audio, PDFs) and a dynamic rendering engine that adapts the UI based on administrator configuration.

Intern (II)

Dec 2025 — Feb 2026

Corporación Z - DatoCohete

Santiago, Chile

- Designed the server architecture for an accounting solution integrated with Chile's Internal Revenue Service (SII) for automated qualification of purchases and taxpayers.
- Implemented omnichannel messaging by integrating the Twilio API, enabling users to manage accounts and clients, access an accounting AI assistant, and obtain legal protection directly from their phone.
- Modeled the database in PostgreSQL with the Prisma ORM on top of a backend built with Fastify and TypeScript.

Software Engineer

Mar 2025 — Dec 2025

Corporación Z - DatoCohete

Santiago, Chile

- Developed multiple web applications based on the Next.js framework and assisted in developing a home-services marketplace mobile app built with Flutter and Firebase.
- Implemented CI/CD pipelines that significantly reduced production deployment times for a proprietary RAG system on Azure.
- Managed deployments on the Vercel hosting platform, configuring DNS on Cloudflare and troubleshooting potential issues.
- Integrated multiple mailing services: configured the Zoho platform for corporate email use and integrated the Resend mailing API for automated email sending. Also configured the DNS records required for proper email delivery.

Full-Stack Dev

Dec 2024 — Feb 2025

Freelance

Santiago, Chile

- Led the technical transition of the platform's checkout, replacing Reveniu with Flow as the payment provider, which required a deep restructuring of the application's core.
- Took advantage of the migration to clean up technical debt and improve database query efficiency (Prisma/MySQL), adapting the system to new operational requirements.
- Ensured the integrity of existing users' data throughout the transition, resulting in zero data loss during the process.

Product Engineer

Aug 2023 — Feb 2024

Freelance

Santiago, Chile

- Led the full lifecycle of a SaaS platform for yoga studios using React, Node.js, and Express. Implemented an automated subscription system with payment integration (via Reveniu) and private areas for users (access to recorded and live classes, among other features).
- Designed a scalable MySQL database with Prisma ORM, optimizing data access. Integrated transactional services with Resend for user management and security.
- Acted as a direct technical consultant, translating client needs into functional requirements through use-case diagrams and prototyping, ensuring delivery of a product aligned with the business.
- Orchestrated deployment on a DigitalOcean VPS, configuring Nginx, Cloudflare, and SSL certificates to ensure high availability and encrypted communications.

Intern (I)

Jan 2023 — Mar 2023

Dyatec

Santiago, Chile

- Administered, supported, and implemented improvements to the GiroLimpio web application.
- Integrated external APIs with GiroLimpio, facilitating data ingestion by users into the platform.
- Migrated the company's mailing system to AWS servers to ensure proper functioning and improve security.
- Held meetings and worked alongside other company interns, guiding them in their tasks and providing support with planning and development of their assigned projects, in order to streamline and optimize development processes within the company.

Back-end Developer

Sep 2022 — Dec 2022

Dyatec

Santiago, Chile

- Designed and implemented the backend (Node.js, Express, JavaScript) and database (MySQL) for a pollutant emission factor tracking system.
- Managed the deployment server configuration (Ubuntu Server) and led the full migration of the system from development to production.
- Held weekly meetings with client company members to report on project progress, review completed work, and verify that objectives and client expectations were being met.

EDUCATION

Universidad de Chile

Santiago, Chile

B.S. in Computer Engineering

Aug 2023 — Dec 2026

- Relevant Coursework: Data Mining; Cybersecurity; Competitive Hacking Workshop.

Universidad Diego Portales

Santiago, Chile

B.S. in Computer and Telecommunications Engineering

Mar 2021 — Jul 2023

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, C/C++, HTML/CSS, Java, Scala, Flutter, Dart, Bash.
- **Technologies:** React, Next.js, Tailwind CSS, Prisma ORM, Git, Django, UNIX, Docker, Jira, NGINX, AWS, Azure, Digital Ocean, Vercel.
- **Databases:** MySQL, PostgreSQL, MongoDB.
- **Tools:** GitHub, Jira, Figma.
- **Languages:** Spanish (native), English (C1).

CERTIFICATIONS

Foundations of Cybersecurity, Google (www.coursera.org/account/accomplishments/verify/EL5DU0GTT0WR) Oct 2025

IBM Z Xplore - Concepts, IBM (www.credly.com/badges/6a545874-76f6-49d2-b726-219e301f13f3) Jul 2024

PROJECTS

Home Lab & Cloud Infrastructure

- Administer a personal server hosting multiple services, including [Plane](#), [n8n](#), and [Authentik](#). Additionally, manage a second server on AWS running a [Pangolin](#) instance, enabling secure access to my services from anywhere via tunneling.
- Configure reverse proxies with the Nginx web server to properly route requests received from Pangolin.

Network Engineering & Hardening

- Designed and implemented a network architecture based on device group segmentation using VLANs, mitigating internal attack vectors and reducing the attack surface. This configuration was carried out using [OPNSense](#) software and managed switches.

EXTRACURRICULAR ACTIVITIES

Career Fair Organization | DCC Universidad de Chile Jul 2024 — Oct 2024

- Responsible for outreach and acquisition of corporate clients for participation in the department's recruitment event.
- Led logistics and internal communication with the DCC to ensure institutional and business objectives were met.

Volunteer | IEEE Computer Society Sep 2023 — Dec 2024

- Collaborated on the logistical organization of the "TechDay" tech event at the FCFM, facilitating networking between academia and industry.
- Supported the management of activities for the student chapter aimed at the community's technical development.